Nemo Outdoor

Introduction

Nemo Outdoor from Keysight Technologies, Inc. has an established position as a pioneer and leader in drive test measurement and optimization solutions for perfecting the air interface of wireless networks. You can gather valuable information for network planning, roll-out, tuning, verification, optimization, maintenance, and benchmarking purposes by collecting measurement results and geographical coordinates. Nemo Outdoor is always among the first-to-market with support for the very latest technologies, currently supporting measurements on 5G NR (SA/NSA) and 2G-4G.

Nemo Outdoor is a centralized solution built on a single, powerful platform. Its expandable, modular structure supports not only drive testing, but also QoS and benchmarking measurements on every standard network technology and on multiple simultaneous data connections. Through its diverse set of options, Nemo Outdoor can be tailored to the specific needs of the customer. This empowers our customers to reach both optimal performance and time- and cost-effectiveness.

Nemo Outdoor is extremely easy to set up, configure and use. Nemo Outdoor‘s customizable user interface offers you the possibility to customize data views and workspaces for any given purpose. Measurement data can be directly imported to various post-processing/analysis tools. In addition, Nemo Outdoor’s import/export feature allows for the transfer of all user settings to or from another computer.

Through our close cooperation with several mobile and scanner manufacturers, we ensure that our customers always get to select amongst the latest test equipment. Currently Nemo Outdoor supports more than 300 test terminals, 5G NR devices and scanning receivers, a figure which is constantly increasing to meet rapidly changing customer needs.

KEYSIGHT

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- Testing with eSIM devices
- Cell locator

Supported Technologies

- 5G NR (SA and NSA)
- 5G NR-DC
- 5G NR carrier aggregation
- 5G mmWave/sub-6 carrier aggregation
- NB-IoT, LTE-M
- LTE/LTE-A (FDD/TDD, carrier aggregation up to 8CC)
- HSPA+ Dual Cell /HSPA+
- GSM
- TETRA
- HSCSD, GPRS, EDGE
- WCDMA
- HSDPA, HSUPA, DC-HSUPA

Key Benefits

- Powerful platform-based tool – All the latest customer needs can be met by adding options to the utterly flexible and expandable Nemo Outdoor platform.
- Comprehensive and cost-effective – Drive testing, Quality of Service (QoS) and benchmarking measurements can all be carried out on a single platform.
- Centralized solution – All of the test devices among the extensive selection can be connected to a single laptop.
- Extremely easy to set up, configure and use – The time from receiving the product to using it is only a matter of a couple of hours. Hence, the customer can focus on the actual task at hand.
- Latest technologies – Nemo Outdoor is always among the first to market with support for the newest technologies.
- Fully customizable user interface – You can tailor the user interface to suit your specific needs at a given time.

Key Features

- Extensive support for 5G NR – Nemo Outdoor supports real 5G NR field measurements with UEs, both Standalone (SA) and Non-Standalone (NSA) modes, and 5G NR carrier aggregation testing, and offers the widest range of 5G NR KPIs on the market.
- Multiple simultaneous data transfers on a single terminal – Nemo Outdoor enables you to perform multiple data transfers on each test mobile in use simultaneously.
- Benchmarking for better performance – Measurements can be carried out on multiple networks and even on multiple technologies simultaneously for ultimate network performance.
- Advanced analytics for easy comparison of terminals, IoT devices and networks
- QoS measurements – Voice quality (POLQA and POLQA 3), YouTube streaming video quality (Keysight's Streaming Experience Quality Inspector algorithm), and video call quality measurements are supported.
- ITU-T **P.863** Perceptual objective listening quality prediction (POLQA version 3) and ViSQOL voice quality measurements.
- Extensive scripting – With Nemo Outdoor's built-in script editor, you can make calls, data transfers, and other tests automatically.
- Measurement file uploading – Measurement files can be sent directly from the Nemo Outdoor user interface to an FTP server, or Nemo Cloud for storage.
- Automatic device detection – Multiple devices can conveniently be added to the system in user-defined order.
- Nemo Outdoor measurement units can be remotely configured and controlled with Nemo Cloud. Nemo Cloud allows a single user to remotely operate Nemo Outdoor test units in the field, tracking their location and status.
- Indoor option – It is also possible to use Nemo Outdoor with the Indoor option on a Tablet PC, and to view the results on an indoor map along with base stations. Also, iBwave floorplans with BTS sites are supported, including real-time small cell and DAS anomaly analysis.
- Instant playback functionality – See the measurement results immediately after a measurement through Nemo Outdoor's playback function and user-configurable data views and workspace.
- Grids, graphs, and map windows can be colored based on user-defined criteria. Graphs support multiple layers for viewing results in a single graph. Another example of the customizability of Nemo Outdoor's data views saving measurement route, or route plan, for later use, and to follow the measurement route through waypoints in real time during measurements.
- Extensive application testing features, including YouTube, Facebook, Instagram, WhatsApp, Viber, Line, Zalo, Snapchat, FAST.com, Bip Messenger, Signal, Telegram any application testing with touchscreen simulation , screencasting, email, MMS, SMS, and voice call testing.
- Emergency call testing for logging the location of the test UE with different positioning techniques
- LTE-A carrier aggregation testing up to 8CC
- User-defined parameters from signaling messages can be searched and displayed in info view and graph side panel.
- Forcing features – Channel/cell (BSIC/scrambling code/PCI) locking, band locking, handover control, timeslot testing, airplane mode, cell barring, carrier aggregation forcing, and AMR codec mode forcing
- Cell testing – The surrounding cells of a location can be tested through an automated list of test calls that are locked to a cell at a time.
- Advanced troubleshooting tools – Missing neighbor detection, pilot pollution analysis, GSM interference analysis

- Mobile band scan – Nemo Outdoor continuously scans for pilot channels from specified bands and automatically adds them for pilot scanning.
- Google Earth map – Nemo Outdoor maps can be exported to Google Earth maps.
- Automated location mapping and automatic indoor positioning features enable displaying the actual test route and location when performing indoor measurement.
- eSIM testing allows you to activate, manage, and switch between multiple eSIMs remotely on all test devices in the field.
- Cell locator tool estimates the physical locations of GSM, WCDMA, LTE, and 5G base stations in real time during drive tests or when playing back measurements.

Nemo Outdoor Supports

- e-mail (SMTP, POP3, and IMAP protocols)
- SMS/MMS
- USSD
- Iperf for UDP/TCP testing
- Ping trace route testing
- FTP/SFTP
- HTTP
- HTTPS file transfers
- HTTP/HTTPS browsing
- Keysight's Streaming Experience Quality Inspector algorithm for measuring real consumer QoE for YouTube streaming
- Video streaming
- Video call quality
- Voice quality (ITU-T P.862.1, ITU-T P.863, WB-AMR, P862.2)
- TWAMP
- ICMP ping
- Facebook, Instagram, WhatsApp, Viber, Line, Zalo, Snapchat, BiP Messenger, FAST.com, Signal, Telegram
- Application Testing Automation (ATA)
- Digital VQ for WhatsApp, Line, Signal, Telegram

Working with Nemo Outdoor

Nemo Outdoor is a comprehensive solution offering you a broad selection of options all on the same platform. Nemo Outdoor has been designed to offer you a pleasant experience with a highly intuitive user interface. The Nemo Outdoor measurement system consists of test mobiles and/or scanning receivers, a GPS receiver with antennas, a PC or a tablet with the Windows operating system, the Nemo Outdoor measurement software, and connecting cables. With the Nemo benchmarking solutions - Nemo Intelligent Device Interface, Nemo Backpack Pro - it is possible to connect 8-24 devices to the test system. With Nemo Network Benchmarking Solution, there are up to 60 devices.

Hardware and Software Requirements

- PC (Lenovo or Dell recommended) with, Windows 10, or Windows 11
- Processor Intel Core i7 2.66 GHz or higher
- SSD drive recommended
- 500 MB of free hard disk space for installation and use; 4 GB recommended
- One USB port for copy protection module
- One USB 3.0 port for test device
- Ethernet port RJ45
- Display resolution 1400 x 900 recommended
- Memory 8 GB RAM minimum, 16GB recommended
- If the Outdoor kit has 20 or more test devices; recommend the PC to have 32 GB RAM
- PDF reader for opening the user manual
- Depending on the scanner used, one USB port or serial port or RJ45 or FireWire port per scanner
- One USB port for an external GPS receiver
- Internal 3G/ 4G modem recommended in case end to end call communication with Nemo Server and Nemo Cloud is used

5G Measurement Solution

5G deployments have begun and rollout announcements from mobile network operators are multiplying. Mobile network operators around the world are investing in 5G and launching networks to capture new business models. 5G introduces new technologies and performance improvements that challenge the way engineers test the network.

Nemo Outdoor supports real 5G NR field measurements with UEs, both Standalone (SA) and Non-Standalone (NSA) modes and offers the widest range of 5G NR KPIs on the market. The solution combines Nemo Outdoor with Qualcomm (X35 RedCap, X50, X55, X60, X65, X70, X75,X80, X85), Samsung Exynos (5100, 5123, 5123A, 5123B, 5133, 5153, 5300,5400, 5410), Balong 5000, and MediaTek (M60 RedCap, M70, M80,M90) modem-based devices and 3rd party scanning receivers.

The collected metrics include cell measurements, physical channel information, current cell information (for each 5G subcarrier), extensive set of RACH parameters, MAC layer KPIs, RLC/PDCP KPIs, and link adaptation (for each 5G subcarrier) KPIs. In addition, Nemo Outdoor collects 5G QoS measurements including throughput, and latency.

When measuring with scanners, the test system can be used to perform real 5G NR measurements with demodulation of the 5G NR reference signals and it enables simultaneous 2G, 3G, 4G, and 5G NR spectrum scan and frequency scan (CW) measurements.

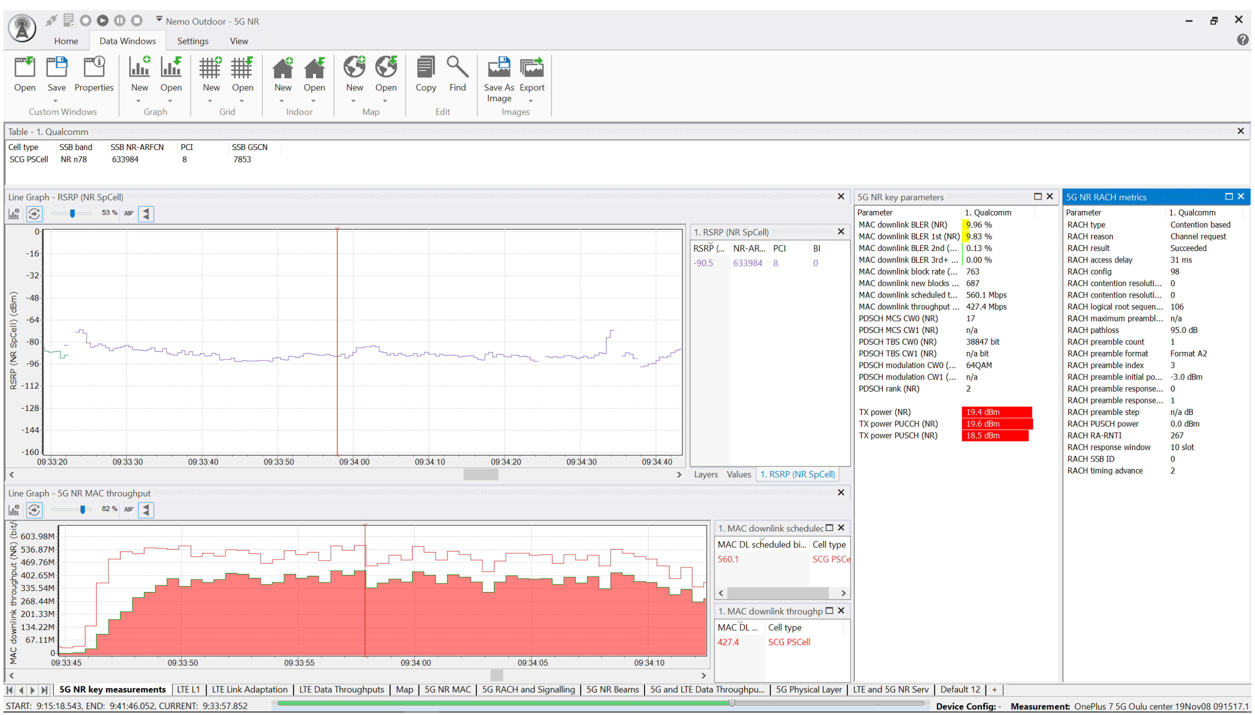


Figure 1. 5G NR views in Nemo Outdoor

5G Features

- Supports Qualcomm (X35 RedCap, X50, X55, X60, X65, X70, X75,X80, X85), Samsung Exynos (5100, 5123, 5123A, 5123B, 5133, 5153, 5300, 5328,5400, 5410), Balong 5000, and MediaTek (M60 RedCap, M70, M80,M90) modem-based devices, and 3rd party scanning receivers
- Supports 5G NR Standalone (SA), Non-Standalone (NSA) and NR-DC mode testing
- Supports 5G NR SA KPIs with Qualcomm, Samsung Exynos, Balong 5000, and MediaTek modem-based devices
- Supports 5G NR SA/NSA forcing and channel/PCI forcing
- Supports 5G NR and 5G mmWave/sub-6 carrier aggregation testing
- Collects 5G NR beam specific KPIs
- Supports massive MIMO (mMIMO) drive tests and field measurements
- Supports Dynamic Spectrum Sharing (DSS) measurements

Dynamic Spectrum Sharing (DSS) provides several advantages to Mobile Network Operators (MNOs), the main one being the opportunity to expand 5G coverage without having to permanently re-farm LTE spectrum or buy 5G spectrum. With DSS, 5G NR network rollout is possible with just a software upgrade, which is not only cost-effective but also quicker than other methods. 5G NR coverage can be provided on existing LTE bands without switching off LTE making it much easier for operators to transition to 5G NR.

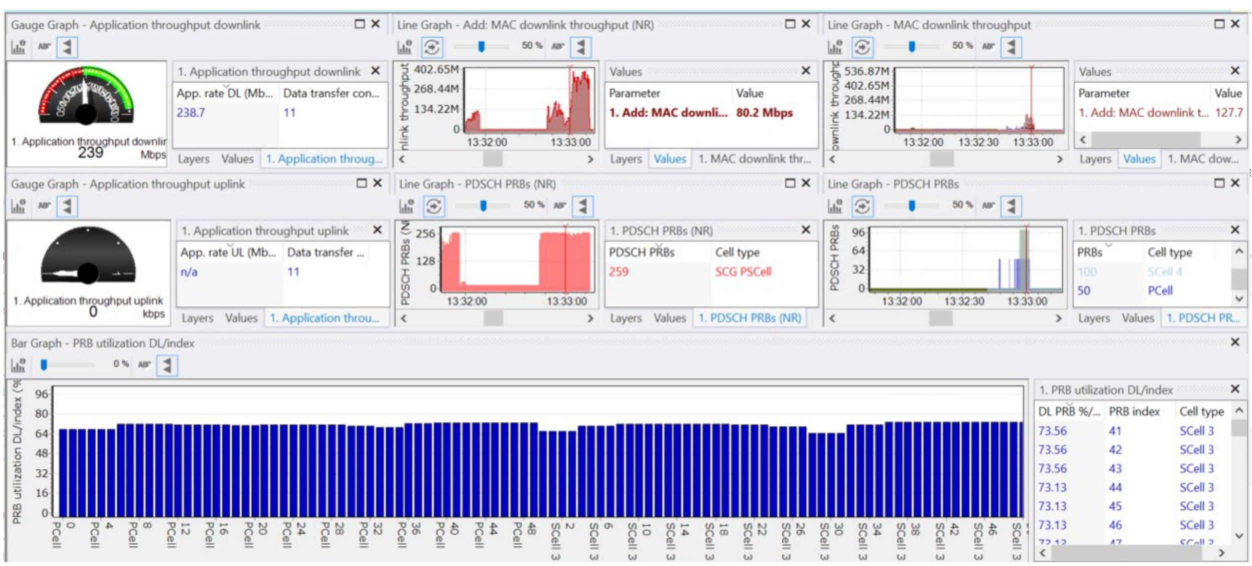


Figure 2. Dynamic spectrum scanning testing in Nemo Outdoor

However, DSS overhead will reduce network capacity. The best way to ensure DSS fits the purpose is to perform lab and field measurements, comparing 5G NR with and without DSS. Nemo Outdoor enables several end-to-end test cases with UEs and frequency scanners, such as:

- Spectrum sharing between LTE and NR to check that all devices, both NR and LTE, get the spectrum allocated in a fair manner when needed
- Total spectrum utilization (spectral efficiency) to verify that the overhead caused by DSS is not more than expected, as per 3GPP specifications
- Mobility and handovers to check how SS-RSRP and RSRP measurements performed by the devices are impacted by changed reference channel allocations
- VoNR and VoLTE to test end-to-end QoS with voice quality measurements

The key metrics to monitor in the DSS test cases are:

- L1 or MAC throughput
- PRB allocations
- Total PRB allocation
- SS-RSRP, RSRP

Application Testing

Test Your Networks and Verify End-User Experience

Nemo Outdoor enables network operators to test their network using the same application protocols as their customers and, therefore, provides results that correspond with the end-user experience. The information provided by Nemo Outdoor assists in the verification and troubleshooting of new services reducing the time-to-market.

Perform application tests either manually or automatically by taking advantage of user-definable scripts or the unique Application Testing Automation method. For all protocols, key performance indicators, such as data throughput, access time, and success rate can be recorded simultaneously with full radio network information.

All services are supported in a single product, Nemo Outdoor, providing an easy-to-use, consistent interface for configuring different applications. Different applications, such as voice, SMS and packet-switched data can be launched in one measurement session to simulate end-user behavior.

Nemo Outdoor supports single and multi (parallel) SFTP transfers and single, multi (parallel) and multithreaded FTP transfers. In single SFTP/FTP transfer, one file is either sent or received. In multi SFTP/FTP transfer, one or more files are either sent or received or both simultaneously. In multithreaded FTP transfer, the transferred file is split into smaller fragments. You define the number of threads. Each thread creates a new FTP session and transfers a fragment of the file. Multithreaded FTP transfer mode is supported in the receive direction.

Social Networking

Facebook, WhatsApp, Instagram, Snapchat, Line, Zalo, BiP Messenger, Signal, Telegram, Viber are some of the most popular social networking applications, and they produce significant amounts of traffic on cellular networks. With Nemo Outdoor it is possible to test the most common functions with these applications and the logged parameters include connection and transfer attempt/success/failure statistics and success rates. Also digital voice quality testing for WhatsApp, Line, Signal and Telegram calls is available.

Nemo Active Testing Application

Nemo Active Testing Application (NATA) is Keysight's proprietary communications interface and application for smartphones. NATA supports on-device measurements with both rooted and non-rooted devices. With NATA, test applications are running on device, hence the test results reflect real user experience. With the NATA interface, smartphones communicate with PC-based applications, such as Nemo Outdoor, enabling voice quality measurements, data transfers, video streaming, and application testing on smartphones without any additional hardware.

NATA is cost-effective and ideal for benchmarking and running several devices simultaneously on a single commercial laptop. As the measurements are running on a smartphone, there are no issues with PC CPU load lowering the data rates. Refer to the NATA data sheet for more detailed information.

Application Testing Automation (ATA)

The Application Testing Automation (ATA) method is based on application-specific test scripts which are used to log pre-defined events for further analysis. With ATA, mobile operators can reliably and rapidly optimize 5G network performance and deliver greater Quality of Experience (QoE) for smartphone users accessing world's most widely used Over-the-Top (OTT) services and social media applications. With ATA, you can test Facebook Messenger, Instagram, TikTok, MS Teams, Snapchat, Zoom, WeChat, Webex, Signal, Telegram, Botim, Google Meet, X(Twitter) and Agnet Work.

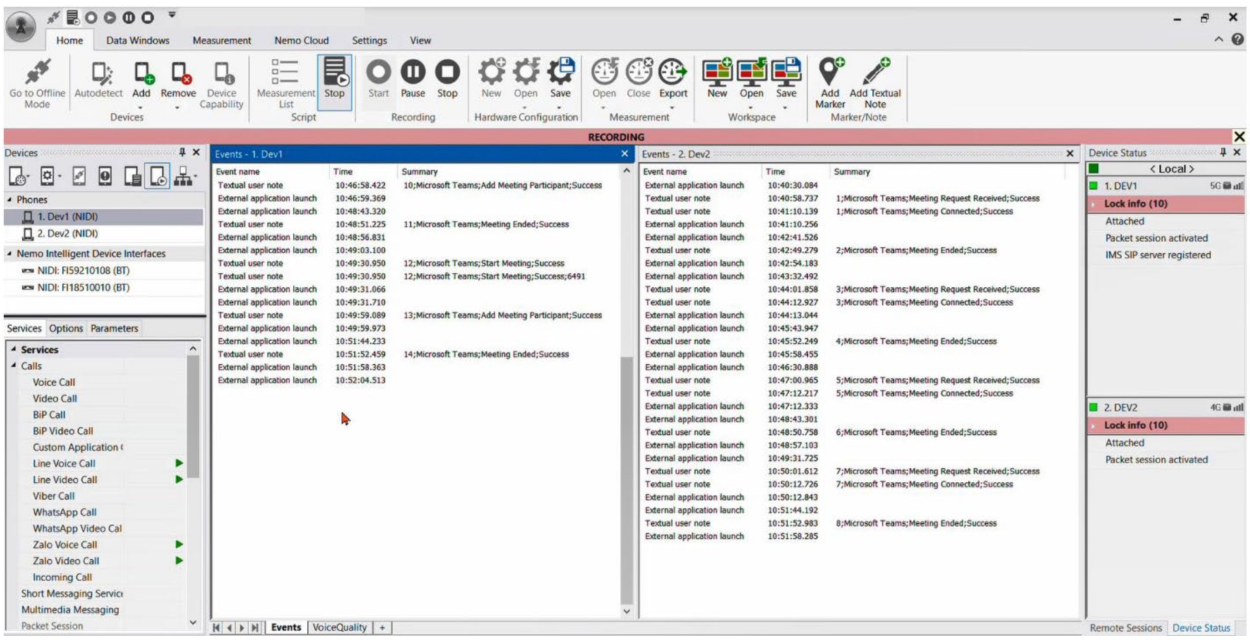


Figure 3. Example view of MS Teams event grids for voice call testing using Application Testing Automation in Nemo Outdoor

Voice and Video Quality

Keysight's Streaming Experience Quality Inspector is an algorithm for measuring on-device, real consumer QoE for YouTube streaming. The algorithm does not require a reference file and it provides accurate scientific MOS results for YouTube video streaming across 2G-5G networks. The license for the new algorithm is Nemo Outdoor platform-based (i.e., not per test device) and it will be available as a USB-portable and as a floating license. This will enable you to use the same cost-effective solution for a high number of test devices and different device models over the years.

IMS Test Features

VoWiFi and ViWiFi Testing

WiFi offload offers a new way for operators to increase the coverage and capacity of LTE networks. WiFi hotspots are available almost everywhere at low cost and all new smartphones include WiFi support. Especially in high-traffic areas, such as shopping malls and sports arenas, WiFi coverage is usually available.

With VoWiFi (voice over WiFi) and ViWiFi (video over WiFi) voice and video information can be sent over the wireless broadband network in digital format. The technology uses the IEEE 802.11 set of specifications for transporting data over wireless local area networks and the internet.

With Nemo Outdoor customers can test the interoperability of LTE and WiFi networks for voice and video call services using Qualcomm and Samsung chipset-based devices. Detailed information about the main Key Performance Indicators (KPIs) and trigger points are logged, including WiFi serving cell information, WiFi SIP registration events and statistics, call events and call statistics, such as WiFi voice call success rate, LTE to WiFi and WiFi to LTE interworking information.

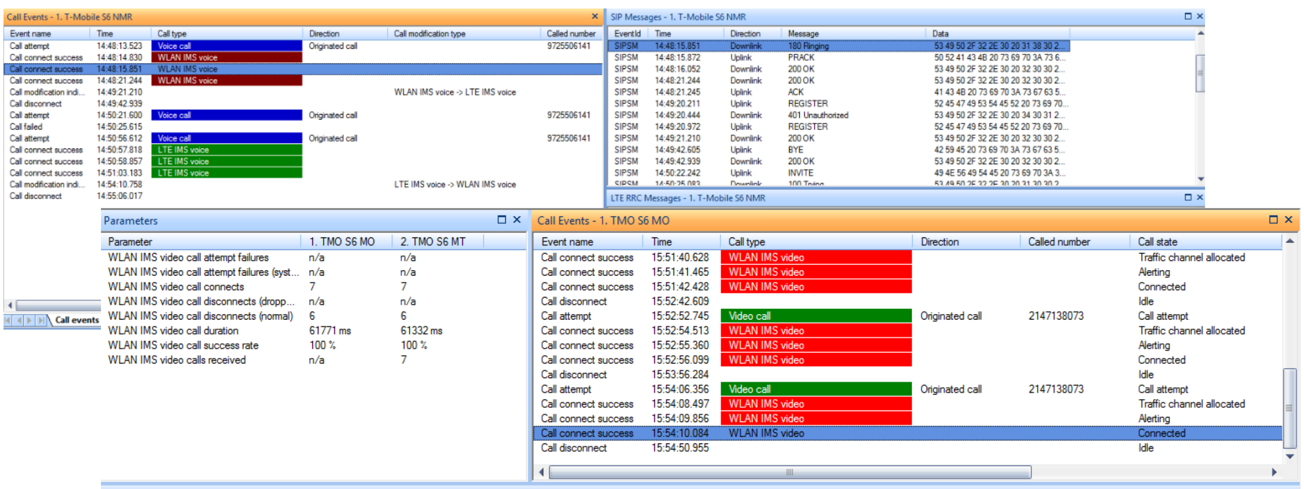


Figure 4. VoWiFi and ViWiFi call events, SIP and RRC messages

VoLTE/ViLTE and SMS over LTE Testing

IMS Profile for Conversational Video Service (IR.94) and IMS Profile for SMS (IR.92) detail how video and SMS are sent in 4G/LTE. Nemo Outdoor collects all the key metrics of VoLTE video calls, including accessibility, retainability, SIP signaling, and RTP packets for video. Also, SRVCC information, including information about the call modification type and result, is logged. VoLTE SMS testing provides the SMS message events for send/receive and also SIP messages. MSG events include the new IMS SMS message type. Video and SMS over IMS is supported with Qualcomm IMS and Samsung IMS clients.

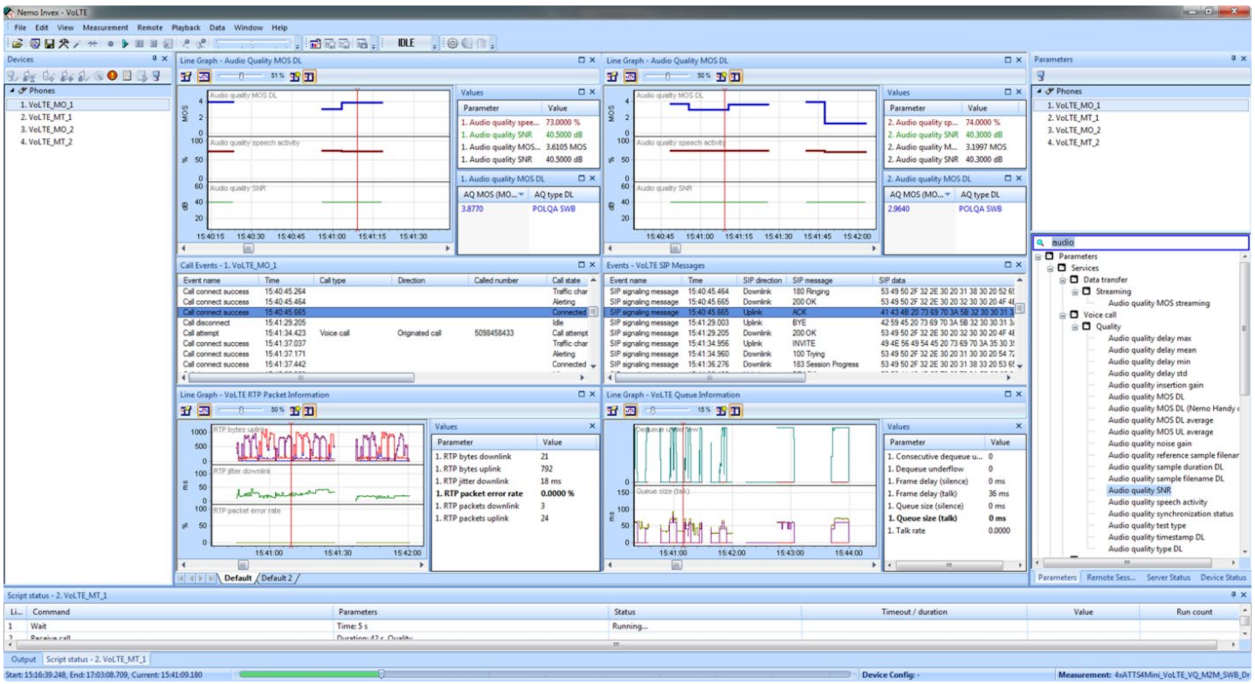
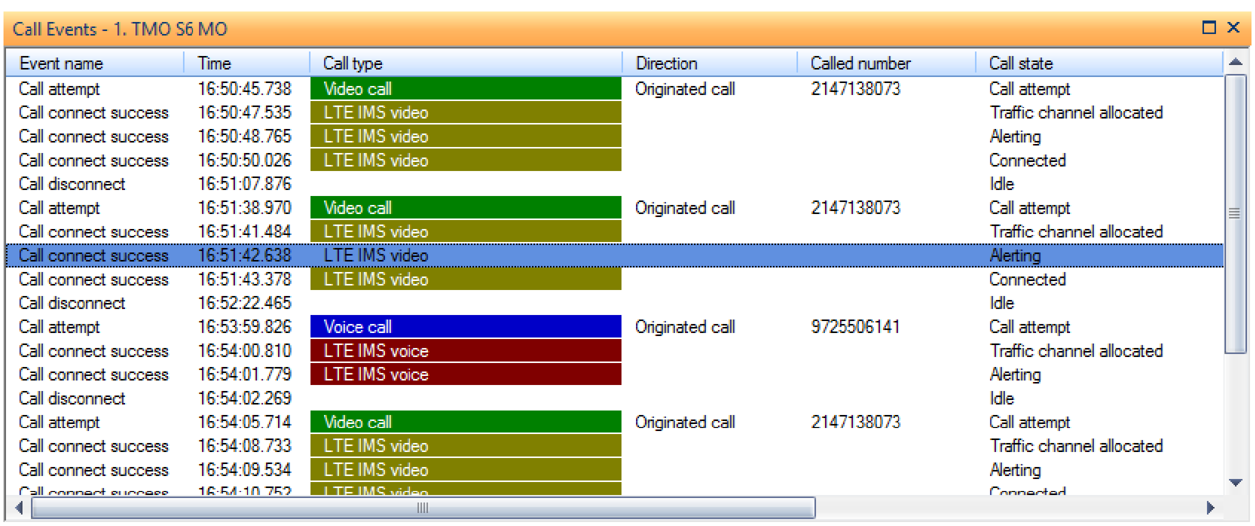


Figure 5. 4 x VoLTE/POLQA measurements



Call Events - 1. TMO S6 MO

Event name	Time	Call type	Direction	Called number	Call state
Call attempt	16:50:45.738	Video call	Originated call	2147138073	Call attempt
Call connect success	16:50:47.535	LTE IMS video			Traffic channel allocated
Call connect success	16:50:48.765	LTE IMS video			Alerting
Call connect success	16:50:50.026	LTE IMS video			Connected
Call disconnect	16:51:07.876				Idle
Call attempt	16:51:38.970	Video call	Originated call	2147138073	Call attempt
Call connect success	16:51:41.484	LTE IMS video			Traffic channel allocated
Call connect success	16:51:42.638	LTE IMS video			Alerting
Call connect success	16:51:43.378	LTE IMS video			Connected
Call disconnect	16:52:22.465				Idle
Call attempt	16:53:59.826	Voice call	Originated call	9725506141	Call attempt
Call connect success	16:54:00.810	LTE IMS voice			Traffic channel allocated
Call connect success	16:54:01.779	LTE IMS voice			Alerting
Call disconnect	16:54:02.269				Idle
Call attempt	16:54:05.714	Video call	Originated call	2147138073	Call attempt
Call connect success	16:54:08.733	LTE IMS video			Traffic channel allocated
Call connect success	16:54:09.534	LTE IMS video			Alerting
Call connect success	16:54:10.752	LTE IMS video			Connected

Figure 6. VoLTE and ViLTE call events

IoT Measurement Solution

Cellular IoT (CIoT) is defined as a set of technologies under 3GPP to enable IoT connectivity using the licensed frequencies, co-existing with the legacy cellular broadband technologies, such as LTE, UMTS and GSM. The main CIoT technologies are NB-IoT and LTE Cat-M1.

Cellular IoT performance is heavily dependent on IoT devices, network equipment implementation, interoperability between the equipment, and network design. Field measurements are important for operators and industrial IoT customers to verify and benchmark the performance of IoT solutions.

Nemo Outdoor provides an impressive set of IoT KPIs with easy-to-use advanced analytics capabilities and it has been used by operators in multiple pilots and coverage measurements in IoT networks around the world. With Nemo Outdoor, you can verify, for example, Connected Mode Mobility (CMM) that enables VoLTE calls to continue when an LTE-M device is moving from a cell to another cell. Nemo Outdoor supports NB-IoT and LTE-M testing with devices based on the Neul and Qualcomm chipsets and with scanning receivers from PCTel and Rohde & Schwarz.

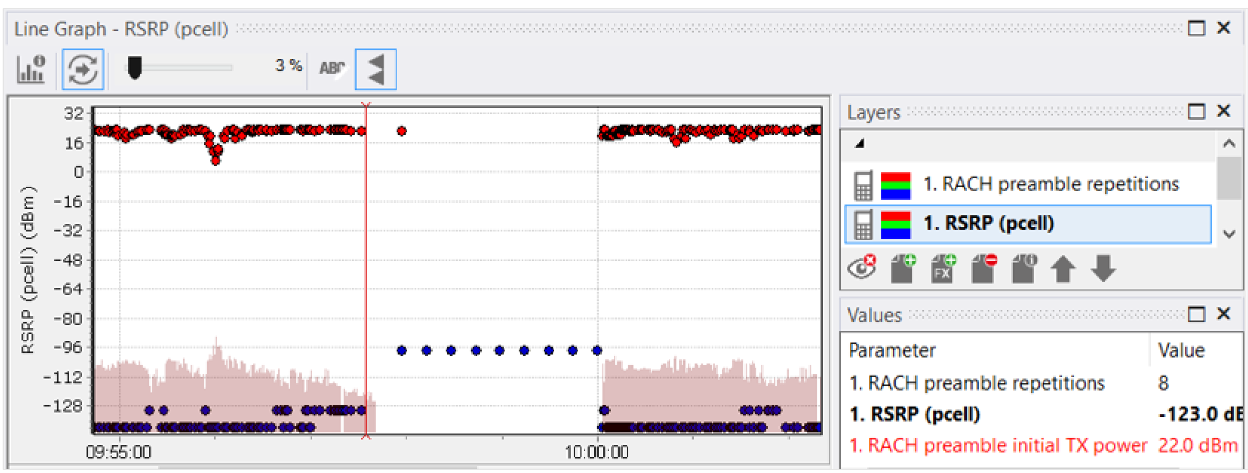
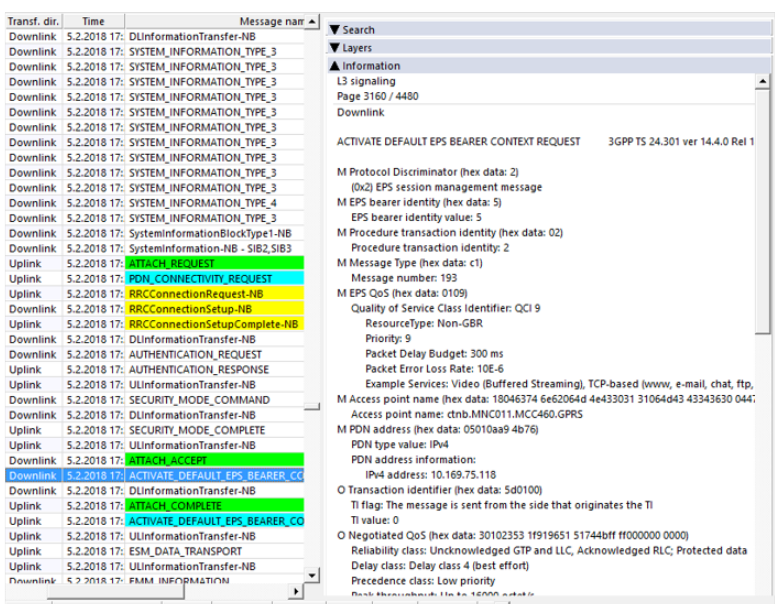


Figure 7. NB-IoT troubleshooting



NB-IoT Features

- Support for Neul and Qualcomm chipsets
- Support for PCTel and Rohde & Schwarz NB-IoT and LTE-M scanners
- Simultaneous LTE/NB-IoT measurements, RSRP, RSRQ, CINR, PSS/SSS CINR with PCTel IBflex scanners
- Support for connected mode mobility (CMM) testing

Supported IoT KPIs

Nemo Outdoor provides an impressive set of IoT KPIs. For example, the following metrics are supported:

- PDSCH/PUSCH throughput
- Ping statistics
- NB-IoT L3 payload UL/DL
- RSRP serving cell avg
- RS SNR serving cell avg
- PRB utilization UL/DL
- TX power PUSCH max/min/avg
- NB-IoT PUSCH subcarriers
- NB-IoT PUSCH/PDSCH repetitions max/min/avg
- RACH access delay
- RACH preamble repetitions
- RACH statistics
- RACH preamble count
- RACH PUSCH power
- RACH pathloss

LTE Test Features

Carrier Aggregation Testing

Staying ahead in a fast-changing competitive environment means being the first to deliver new functionalities and impeccable quality. To better serve our network testing customers, Nemo Outdoor offers support for LTE-A 8CC DL and 3CC UL carrier aggregation testing, with the same data protocols and services used by subscribers.

With Nemo Outdoor operators can monitor the performance and quality of their LTE-A networks. The measurements provide detailed information about the primary and secondary component carriers (SCell1 and SCell2), including cell type, PCI, CFI, band, bandwidth, link adaptation and radio resource allocation.

The examples below show how LTE data is presented in the Nemo Outdoor user interface.

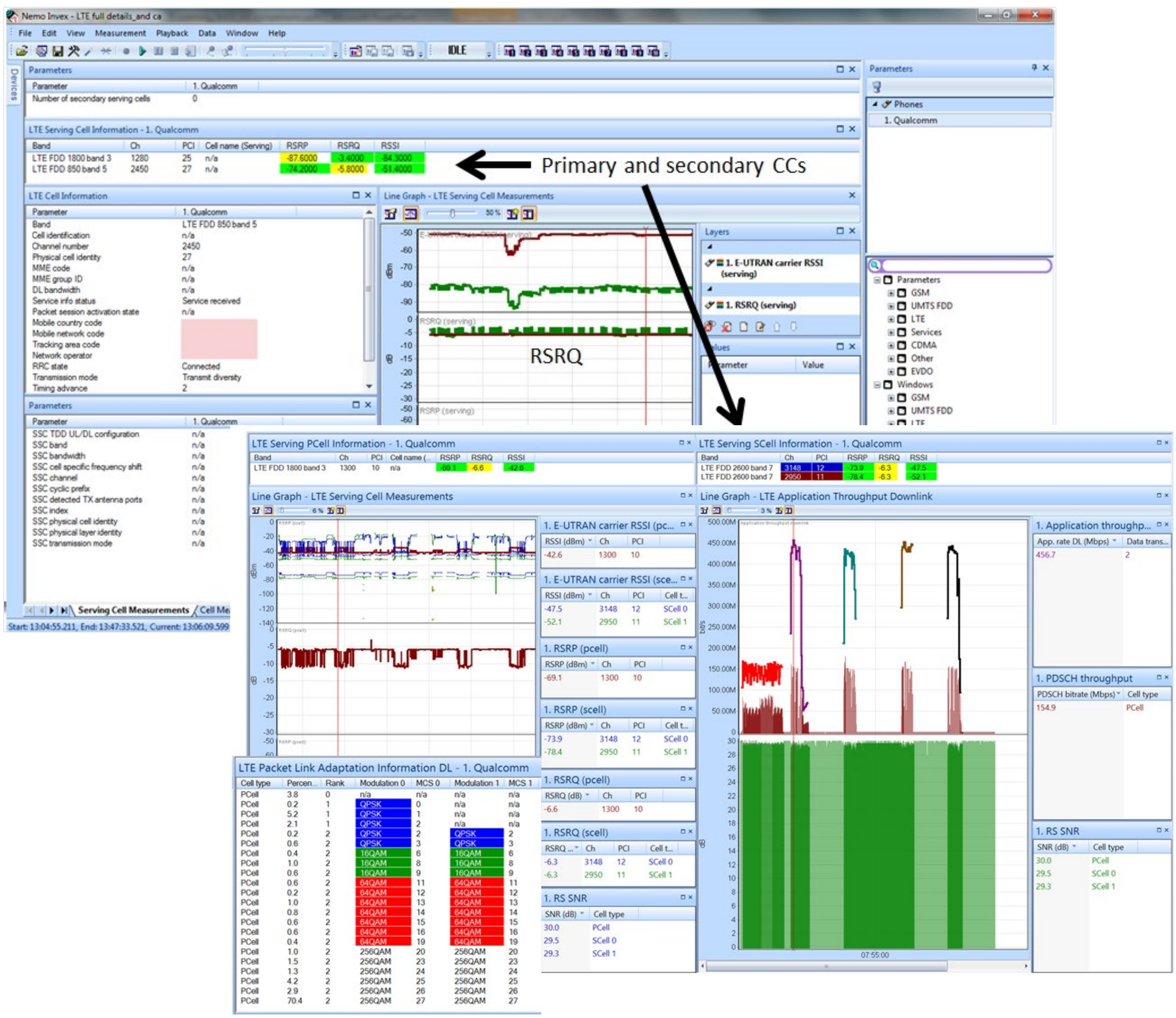


Figure 8. Two LTE carrier components

Multimedia Broadcast Multicast Service (eMBMS)

Video streaming services have become very popular and take up a major share of internet traffic worldwide. LTE broadcast is a Single-Frequency Network (SFN) technology that enables multiple users to receive the same multimedia content, such as live sports, simultaneously using Point-to-Multipoint (PMP) distribution. LTE broadcast is part of the series of 3GPP LTE standards known as evolved Multimedia Broadcast Multicast Service (eMBMS).

With Nemo Outdoor, operators can capture and decode eMBMS information using Qualcomm chipset-based devices, and with scanners from PCTEL and Rohde & Schwarz. Measurement data contains LTE eMBMS active bearer info, signaling messages, and main KPIs, including eMBMS data throughput for different layers (PMCH, RLC and MAC), eMBMS configuration and PMCH information for LTE broadcast services. MBMS related information is logged to MBMS information (MBMSI) and Physical channel throughput broadcast (PHRATEB), and to MAC layer throughput broadcast (MACRATEB) events.

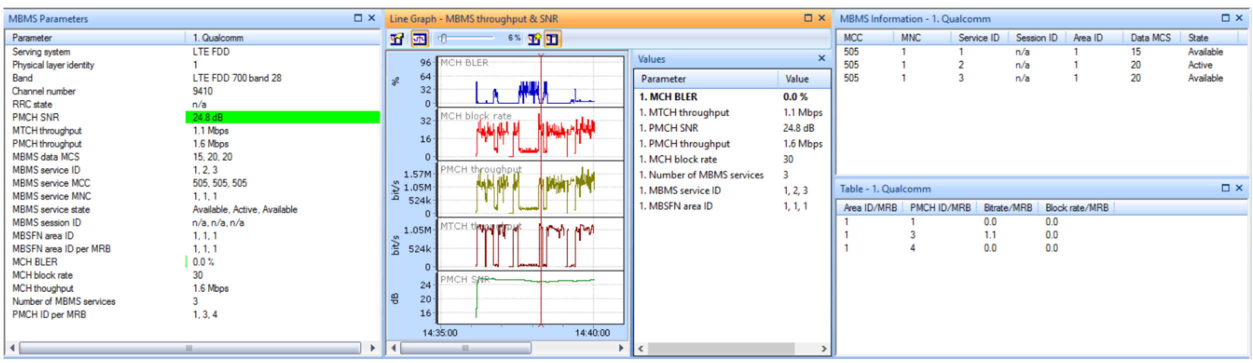


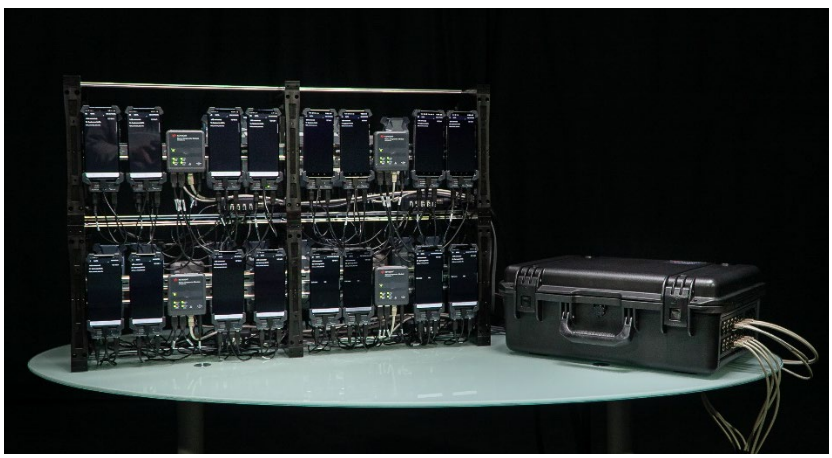
Figure 9. eMBMS information in Nemo Outdoor

Benchmarking

You can rely on Keysight's impressive benchmarking solutions Nemo Network Benchmarking Solution (NBM), Nemo Intelligent Device Interface (NIDI), Nemo Backpack Pro. Benchmarking measurements can be performed on any combination of networks and system technologies ranging from GSM and WCDMA to HSPA+, LTE, and 5G NR.

Depending on the test setup, as many as 60 test devices can be connected with Nemo Outdoor at once as well as scanning receivers from PCTEL and Rohde & Schwarz. In addition, they are all conveniently connected to the same Nemo Outdoor platform running on a single laptop. Furthermore, combined with Nemo Server, you can conduct long-term network performance measurements.

Please refer to the Nemo Network Benchmarking Solution, Nemo Intelligent Device Interface, Nemo Backpack Pro product materials for more detailed information.



Nemo Network Benchmarking Solution (NBM)

Nemo Intelligent Device Interface (NIDI)

Nemo Backpack Pro

Scripts and Measurement Lists

In Nemo Outdoor measurement automation is enhanced through scripting. By creating and editing script files with Nemo Outdoor's built-in script editor, Nemo Outdoor makes calls, data transfers, etc. automatically.

Nemo Outdoor's script group functionality enables you to synchronize script files and/or certain lines in the script file for specific devices. Both synchronization methods can be used together or separately depending on your needs.

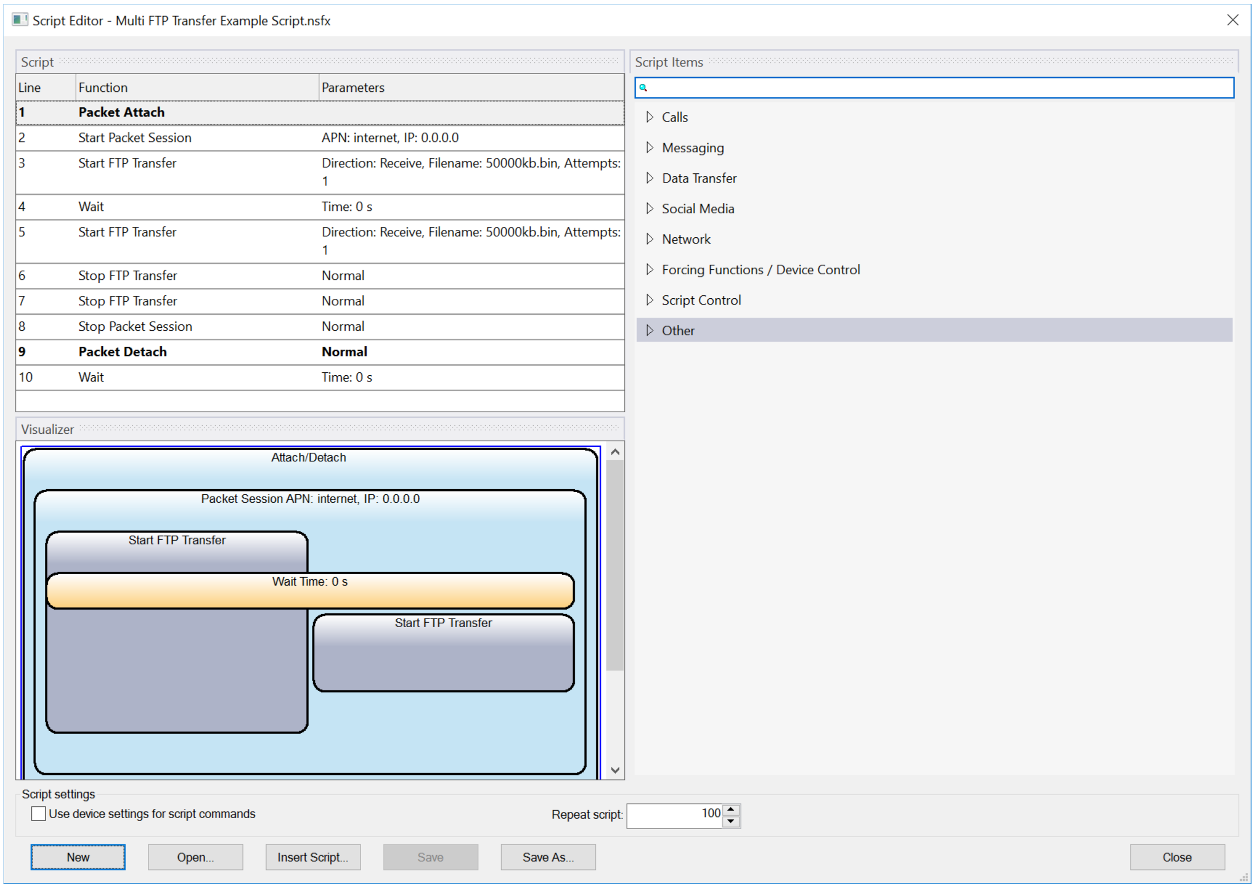


Figure 10. Nemo Outdoor Script Editor

Measurement lists are more complex than individual scripts, enabling even larger-scale measurement campaigns and increased measurement automatization. Measurement lists, which can be loaded and saved, enable you to run automated measurements with several devices combining multiple scripts. What is more, you can use measurement lists with just one device to run several scripts sequentially, one after another. However, each measurement is individually recorded in a separate file. You are also able to use measurement lists without scripts.

Nemo Outdoor Device Setup

Nemo Outdoor has a flexible and user-friendly user interface which allows users with different levels of experience to have an easy and smooth access to the system. The automatic device detection functionality automatically detects devices connected to the PC and assigns them the appropriate COM port and data connection settings. You can determine the order of added devices and can set the devices to start automatically upon setup. You do not need to spend time on manually going through the procedure; let the easy and intuitive Nemo Outdoor user interface automatically handle device connections for you.

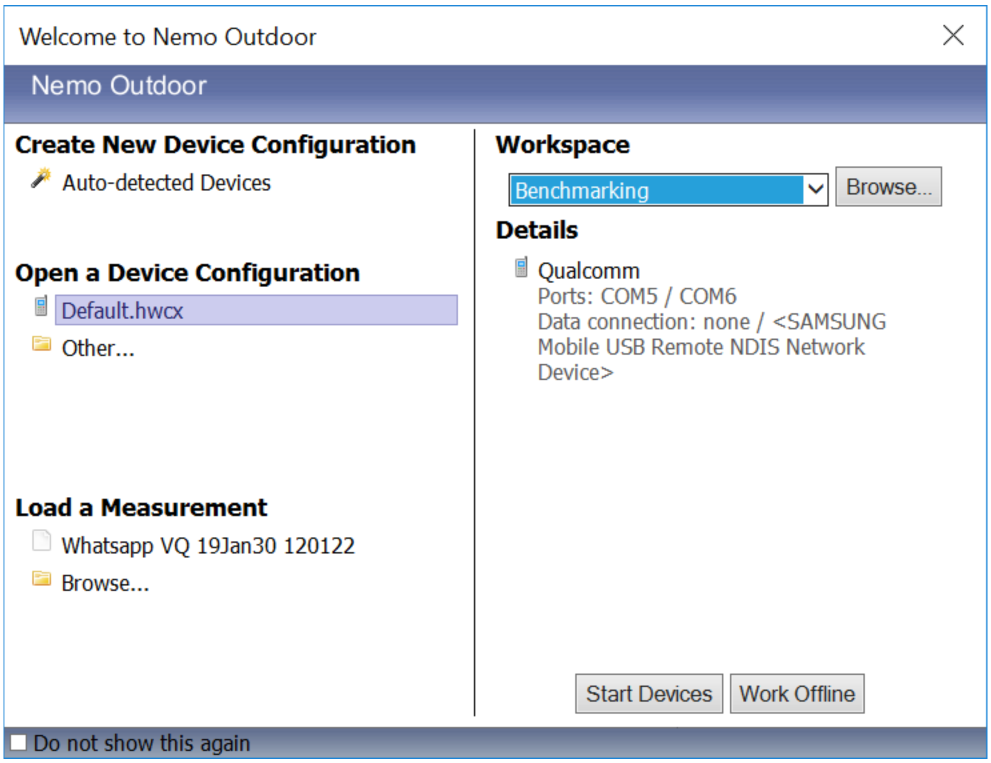


Figure 11. The automatic device detection functionality makes Nemo Outdoor easy to set up and use

RF Validation

In order to prevent measurements errors caused by a faulty antenna or loose RF cables, you can perform RF cross-checking with the test devices connected to Nemo Outdoor. This can be done using the RF offset parameter, RF test feature, level check, or scanner configuration check.

With the RF test feature, you can validate all test devices before starting measurements. The RF test gives detailed information about the RF performance of the connected test devices, UEs and scanners. During the RF test, the average and max average field strength values are calculated for all serving cells measured and then the results reported by all devices are compared.

RF Test Results

Hardware configuration: **Default**

Start: **27.11.2015 13:24:05**

End: **27.11.2015 13:25:05**

Device: **1. Qualcomm**

System	Parameter	Channel	Cell	Average	Max.Avg.	Delta	Status
UMTS FDD	RSCP (UMTS)	10837	7	-68.9025	-68.9025	0	Pass

Failed: 0

Device: **2. Qualcomm**

System	Parameter	Channel	Cell	Average	Max.Avg.	Delta	Status
UMTS FDD	RSCP (UMTS)	2963	1	-62.9174	-62.9174	0	Pass

Failed: 0

Figure 12. RF test results in Nemo Outdoor

With the level check you can compare field strength values measured by a scanner and test UEs. The data is displayed in a table grid. This helps you detect possible antenna and cable defects real-time during measurements and also during playback.

LTE RSRP level check - 1. Qualcomm

Band	Ch	PCI	RSRP	RSRP (ref.)	RSRP (delta)
LTE FDD 1800 band 3	1300	125	-66.3	-61.3	-5.0

GSM RX level check - 1. Qualcomm

Band	ARFCN	BSIC	Cell ID	Cell name (Serving)	RxLev sub	RX level (ref.)	RX level (delta)
GSM 900	39	27	14758	n/a	-53.0	-53.5	0.5

Figure 13. Level check compares field strength values measured by a scanner and test UEs

Another way to prevent measurements errors caused by missing/incorrect RF antenna connections is to use the RF offset check when the measurement is started. The RXL/RSSI value of each measured channel/carrier is read from each connected phone and scanner. The value from each channel is compared to the highest value from all devices. If the measured value is less than X dB (= user-definable RF offset value) below the highest value, a warning message is displayed with the channel number and the device name(s). In Remote mode, Nemo Cloud can set the offset value X. If the field unit triggers a warning message, it is shown in Nemo Cloud.

With the scanner configuration check you are notified during a measurement if a scanner is not configured to scan all serving cell channels measured by the UEs configured in the system.

Centrally Controlled Testing with Nemo Cloud

Nemo Cloud is a one-stop-shop platform for centrally controlled network testing from measurement session creation to final measurement reports. It allows network operators to remotely operate Nemo Outdoor and Nemo Handy/Nemo Handy Autonomous test units and check their status. The location and status of field units, including the driver's contact details, are tracked in Nemo Cloud. Configuration settings can be shared to and between all measurement units within the same project quickly and easily. Nemo Cloud supports all devices and test protocols currently supported with Nemo tools.

A single user can remotely monitor and manage numerous Nemo measurement kits in the field thus eliminating the need for two technicians in a test vehicle. Measurement sessions are created and scheduled in Nemo Cloud and they can be assigned for individual measurement units or to groups of units. Field unit hardware configurations and test device details can be verified remotely to avoid failed drive tests caused by malfunctioning hardware. In addition, hardware failures, incorrect scripts and other issues can be spotted and corrective measures taken immediately instead of at the end of a one-week test campaign. By reacting without delay, both time and money can be saved.

For locally controlled Nemo tools, Nemo Cloud provides file and configuration sharing between measurement units. This reduces work before the start of the measurement project in the field and eliminates human errors in local measurement unit configuration.

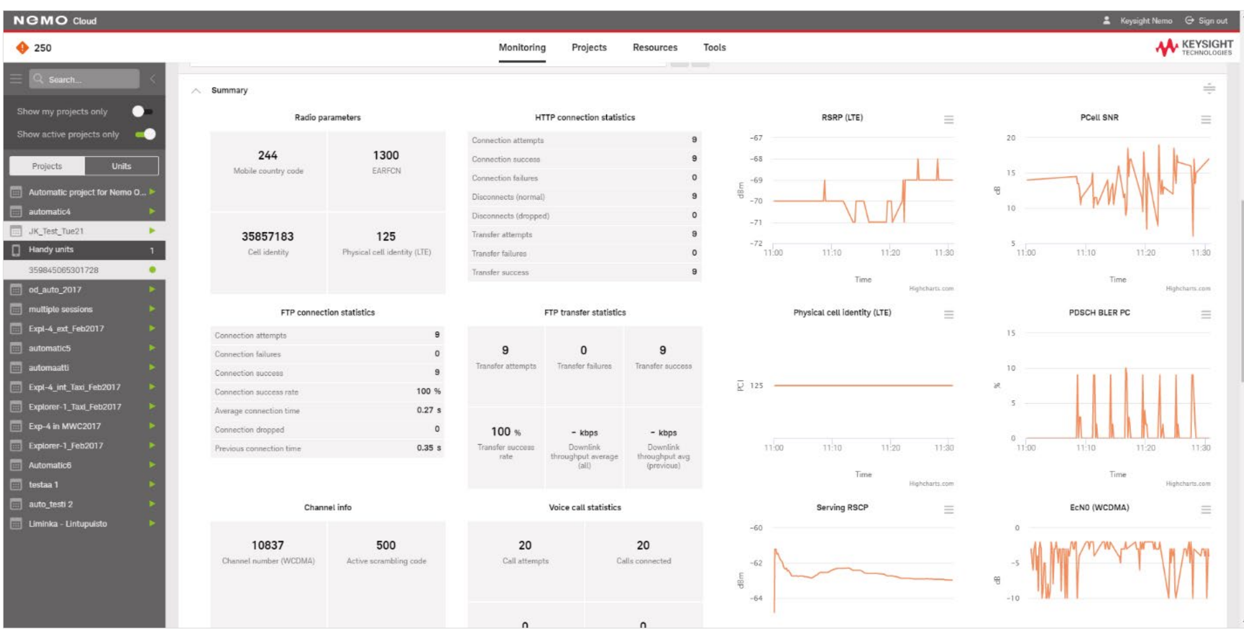
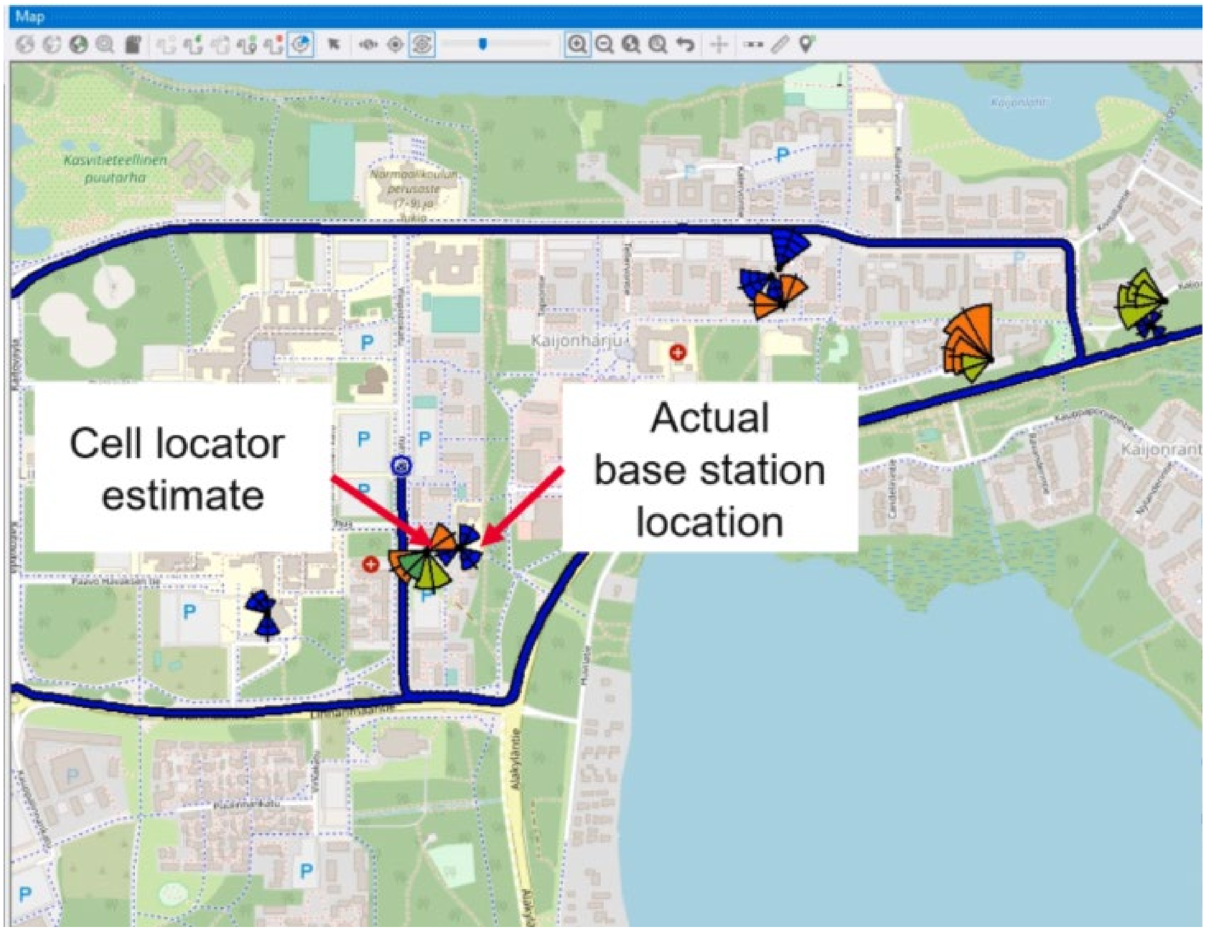


Figure 14. Nemo Cloud provides a centralized platform for viewing information about your measurement projects

Cell Locator

With the cell locator tool, Nemo Outdoor estimates the location of base stations based on signal strength measured during drive tests. When a cell is detected, it is displayed on the map. The estimated location of the cell will gradually refine and move closer to its actual physical location based on the coverage extent of the drive test. The cell locator tool gives the best results when you drive around the entire cell. The cell locator feature works both in recording and playback mode.



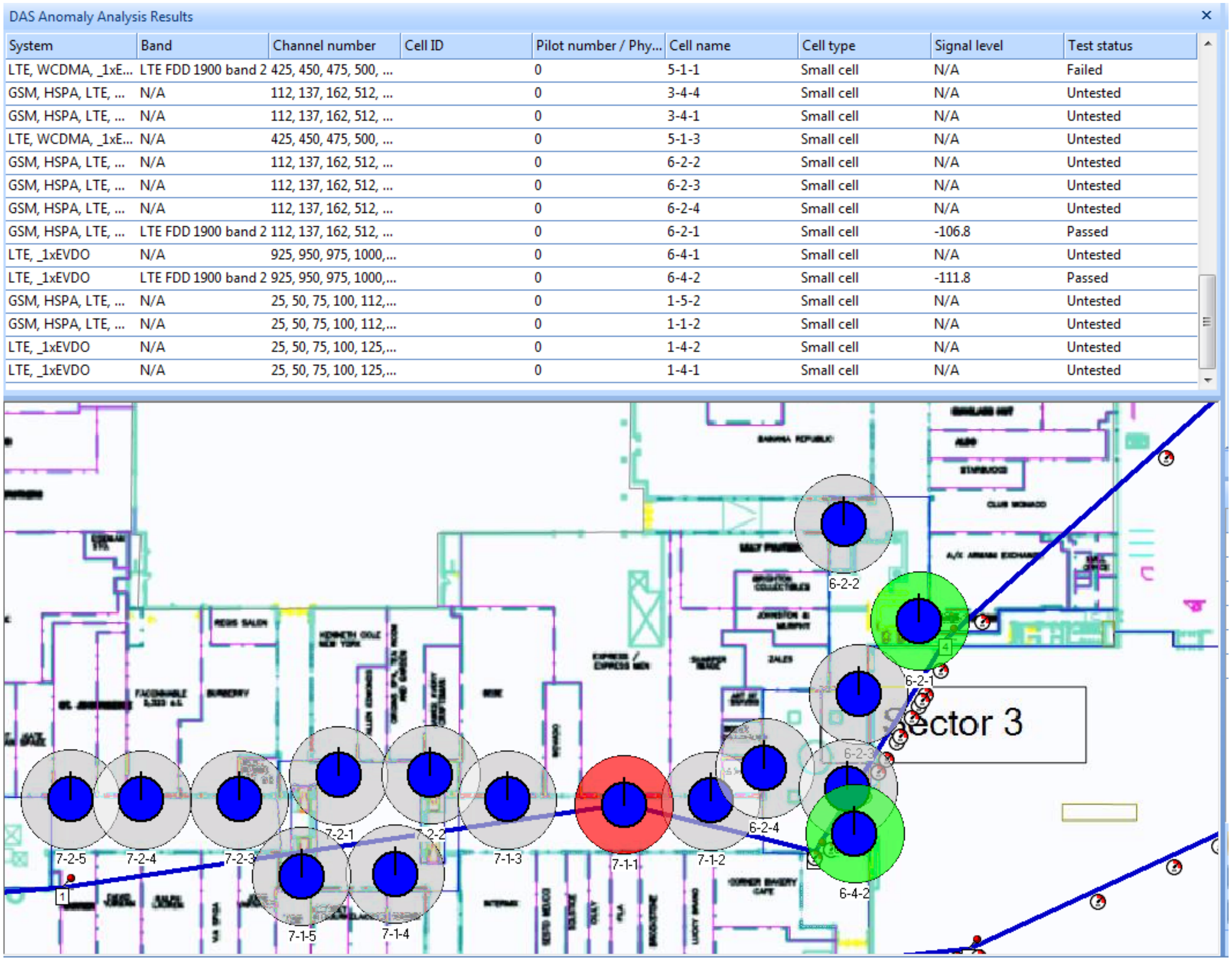
Troubleshooting and Verification

Network Optimization

Networks are in a constant state of evolution, and require continual optimization, troubleshooting, verification, and fine-tuning. Nemo Outdoor offers tools for various troubleshooting and verification tasks, such as DAS anomaly analysis, real-time RF ingress and missing neighbor detection, pilot pollution analysis, and GSM interference analysis.

DAS Anomaly Analysis

With DAS anomaly analysis you can easily check and verify in-building small cells and Distributed Antenna Systems (DAS) antennas by giving a pass, fail, or untested result for each cell. With this simple test, network engineers can quickly verify that all cells are transmitting without the need to revisit the indoor venue. DAS anomaly analysis is supported with GSM, WCDMA, LTE, and WLAN technologies.


DAS Anomaly Analysis Results

System	Band	Channel number	Cell ID	Pilot number / Phy...	Cell name	Cell type	Signal level	Test status
LTE, WCDMA, _1xE...	LTE FDD 1900 band 2	425, 450, 475, 500, ...		0	5-1-1	Small cell	N/A	Failed
GSM, HSPA, LTE, ...	N/A	112, 137, 162, 512, ...		0	3-4-4	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	N/A	112, 137, 162, 512, ...		0	3-4-1	Small cell	N/A	Untested
LTE, WCDMA, _1xE...	N/A	425, 450, 475, 500, ...		0	5-1-3	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	N/A	112, 137, 162, 512, ...		0	6-2-2	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	N/A	112, 137, 162, 512, ...		0	6-2-3	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	N/A	112, 137, 162, 512, ...		0	6-2-4	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	LTE FDD 1900 band 2	112, 137, 162, 512, ...		0	6-2-1	Small cell	-106.8	Passed
LTE, _1xEVDO	N/A	925, 950, 975, 1000,...		0	6-4-1	Small cell	N/A	Untested
LTE, _1xEVDO	LTE FDD 1900 band 2	925, 950, 975, 1000,...		0	6-4-2	Small cell	-111.8	Passed
GSM, HSPA, LTE, ...	N/A	25, 50, 75, 100, 112,...		0	1-5-2	Small cell	N/A	Untested
GSM, HSPA, LTE, ...	N/A	25, 50, 75, 100, 112,...		0	1-1-2	Small cell	N/A	Untested
LTE, _1xEVDO	N/A	25, 50, 75, 100, 125,...		0	1-4-2	Small cell	N/A	Untested
LTE, _1xEVDO	N/A	25, 50, 75, 100, 125,...		0	1-4-1	Small cell	N/A	Untested

Figure 15. DAS anomaly analysis

RF Ingress Detection

RF ingress occurs when outdoor cells (macro cells) interfere with indoor antenna systems in which case immediate network optimization is required. Nemo tools calculate the distribution between the small cell/DAS and macro cell coverage inside a building and give real-time information whether RF ingress by a macro cell is detected or not.

GSM Interference Analysis

In GSM mobile communication networks, system capacity is often limited by co-channel interference. When surrounding cells use the same carrier frequency or use a channel too close to the one used by a terminal, this creates adjacent or co-channel interference. GSM co-channel and adjacent channel analysis is done in real time during a measurement and playback with Nemo Outdoor. Interference detection is done based on GSM terminal and GSM scanner measurements and these measurements are combined for analysis.

Pilot Pollution Analysis

Available for WCDMA networks, pilot pollution analysis measurements can be performed in real time with Nemo Outdoor by using test terminals or a scanning receiver. In a pilot pollution situation, there are more active/monitored pilots than a handset can measure, or there is no clear dominance of a single pilot in the area. The pilot signal is used to distinguish cells in the network from one another. Pilot pollution analysis is done based on cell measurement events, and analysis is always active. You can define thresholds for pilot pollution analysis via the Nemo Outdoor user interface.

Pilot pollution situations detected with WCDMA scanners and terminals, and GSM co-channel/ adjacent channel interference situations can be displayed on a map. A line from the current location is drawn to the interfering cell(s). A BTS file with WCDMA scanners and terminals can be used to display cell names for cells causing pilot pollution.

IP Packet Capturing

With IP packet capturing, network packets sent between IP addresses are stored in log files and can be post-processed with a third-party application such as Ethereal. A separate log file is generated for each test terminal making data transfers. IP packet capturing is especially useful in problems related to TCP/IP communication which do not necessarily appear in the measurement logs.

Data Views and User Interface

Nemo Outdoor’s flexibility is best displayed and experienced through its class-leading user interface. It is arranged into control and data windows which can further be arranged and adjusted depending on your needs.

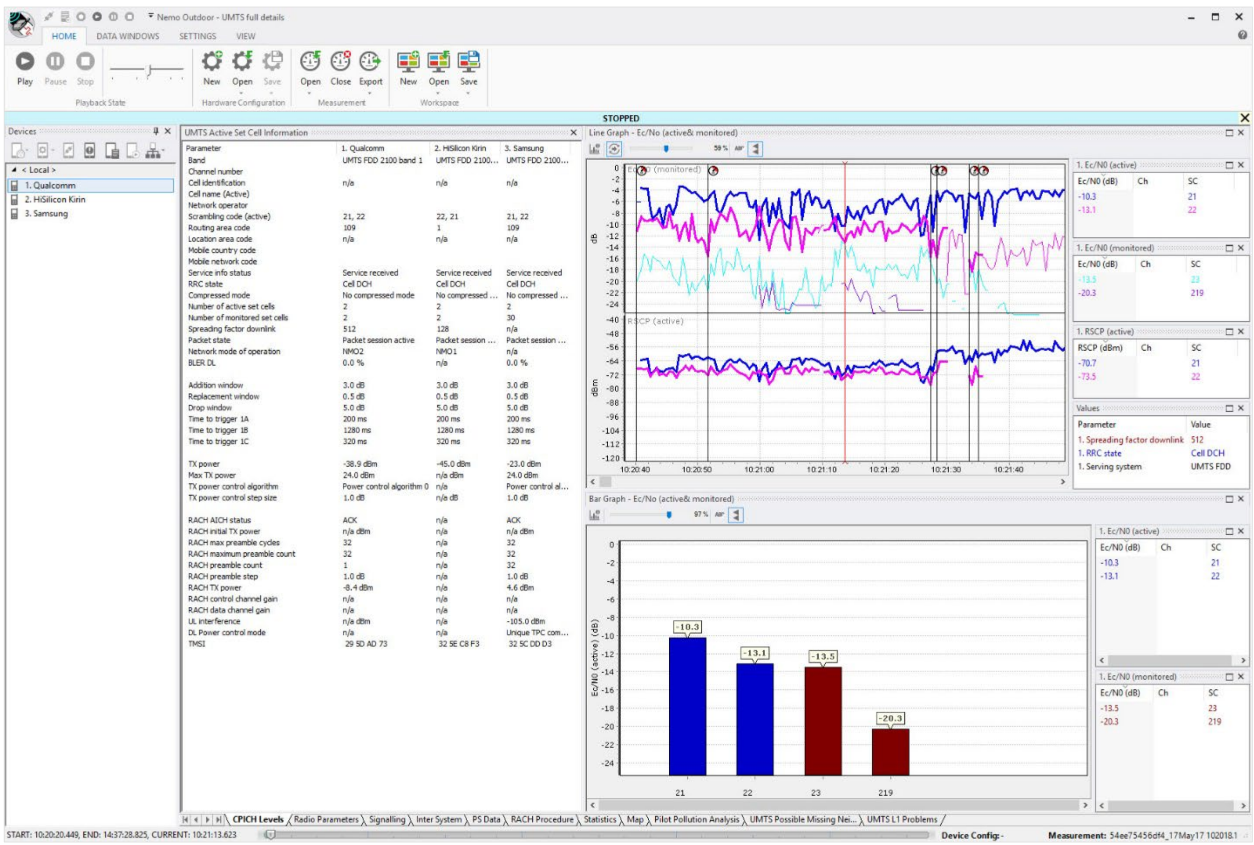


Figure 16. Nemo Outdoor user interface is extremely flexible

The Nemo Outdoor user interface is compatible with common Windows standards. During measurement, you can easily monitor the results and the progress of the measurement process. After the initial device setups and configurations, you can save all device-related settings to a hardware configuration file and load the same configuration later. The Nemo Outdoor user interface is first and foremost flexible and customizable to suit your specific needs. View groups allow the organization of measurement windows into different tabs for easier viewing. This is especially useful when there are several graphs and maps open simultaneously in the Nemo Outdoor main window.

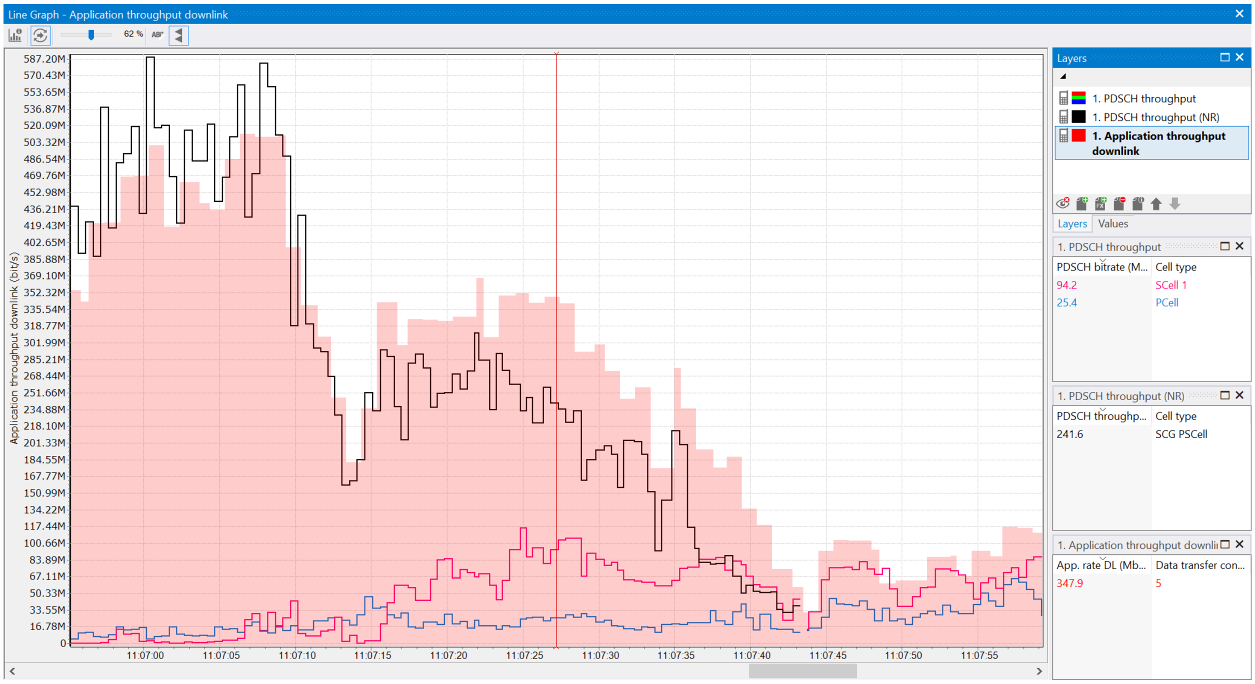


Figure 17. A 5G line graph in Nemo Outdoor

Nemo Outdoor maps provide you with a better understanding of the measurement route and results through visualization. When Nemo Outdoor is used with a GPS receiver and positioning coordinates are collected, the measurement route is drawn on a map and you can easily correlate events to location coordinates. Most of the time drive testing is performed using regular GPS receivers that give exact coordinate information in normal situations. Once the GPS fix is lost, however, coordinates cannot be updated correctly. In areas where there are a lot of tunnels or other obstacles, the loss of GPS signal can affect the accuracy of your network measurements and negatively impact your test results. With Nemo Outdoor you can correct measurement route drawings, for example, when the route is missing because the test vehicle drove through a tunnel.

Nemo Outdoor offers parameter-based route coloring, which means that you can observe the values of certain network parameters from the route coloring on the map. You can define which color refer to which parameter value. This way it is easy to spot the problem areas on a map. To make analysis even simpler, the same route can be drawn several times on the map and different route coloring can be applied to each of them. In addition, certain events can be shown as icons on the map.

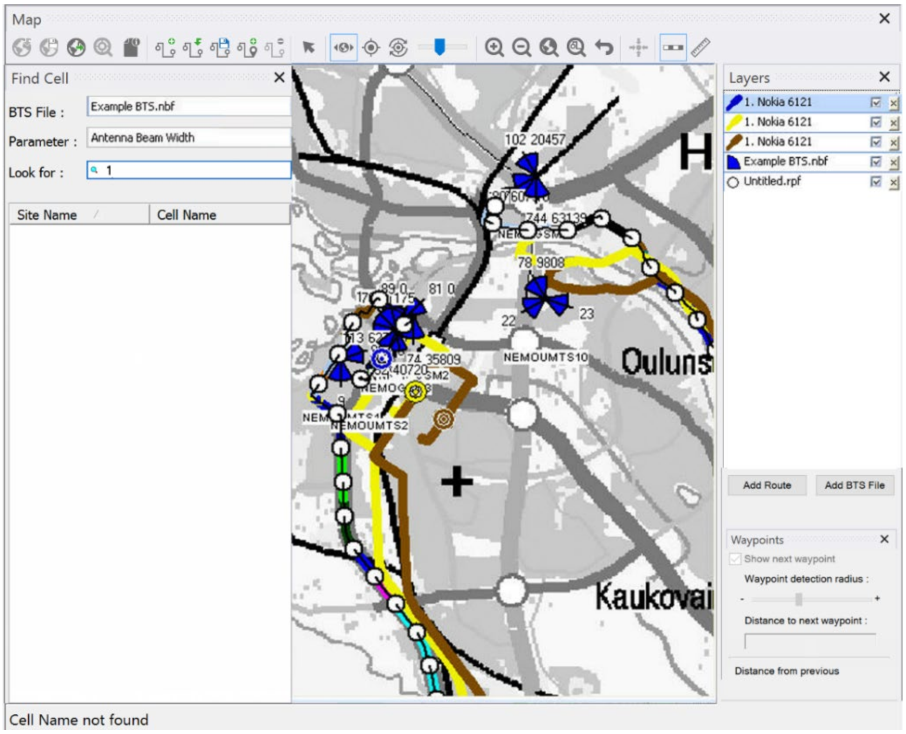


Figure 18. On a map you can view, for example, base station icons and a route plan

Nemo Outdoor maps and floor plans can also display a base station overlay. With a user-defined BTS file, the map shows the location of each base station, the defined antennas with antenna directions, and even antenna apertures and cell identifying parameters. Furthermore, all this information can be edited directly in Nemo Outdoor. During drive testing or playback, a line connecting the current location to the serving and/or neighbor cell (sector) will be drawn automatically. This provides a highly visual impression of the network operation. For example, if a call is connected to a non-optimal cell, it will appear instantly on the map. The map will also display missing neighbors for quick identification and resolution.

Nemo Outdoor supports Google Map, MapInfo, OpenStreetMap, and iBwave map formats. It also supports MapXtreme Geoset files (.gst) and GPS eXhange format (GPX). All user-defined map settings, such as, the order of the different map layers and the zoom factor are stored in the .gst file. Nemo Outdoor maps can also be exported to Google Earth maps. Furthermore, Google Earth can be used to create KMZ files, georegistered raster images of floorplans, which can be used as indoor maps in Nemo Outdoor.

Decoding

In events and messages grids, the data can be analyzed even more in-depth by decoding the individual events and messages simply by double-clicking the event in question. You can define both the background and text color in event and message grids. This color coding can be done based on a certain message, sub channel or decoded message.

Notifications

Notifications enable you to add another dimension to the measurement process. Audio prompts help you during drive testing to immediately notice when something special happens. You can configure Nemo Outdoor to play audio notifications or voice prompts whenever a certain event occurs. The voice prompts are Windows .wav files, and in Nemo Outdoor both female and male default notification sounds are available. If you prefer, you can record and use your own voice notifications instead of the default files. Custom notifications can also prove useful when creating a script file. For example, you can set a video call to start only when it is supported by a cellular technology. Nemo Outdoor's advanced exporting functionality also enables you to export individual custom-made notifications.